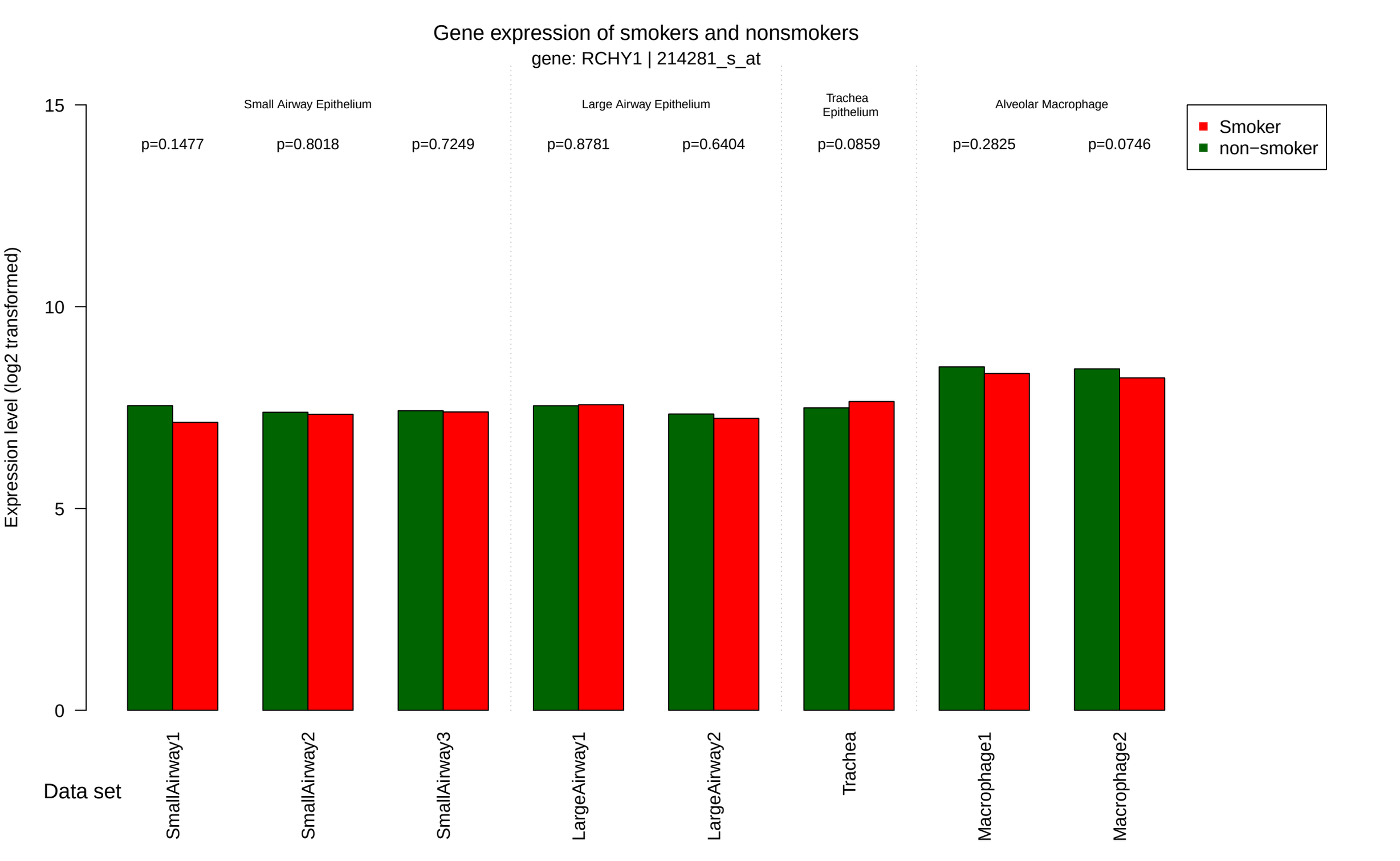
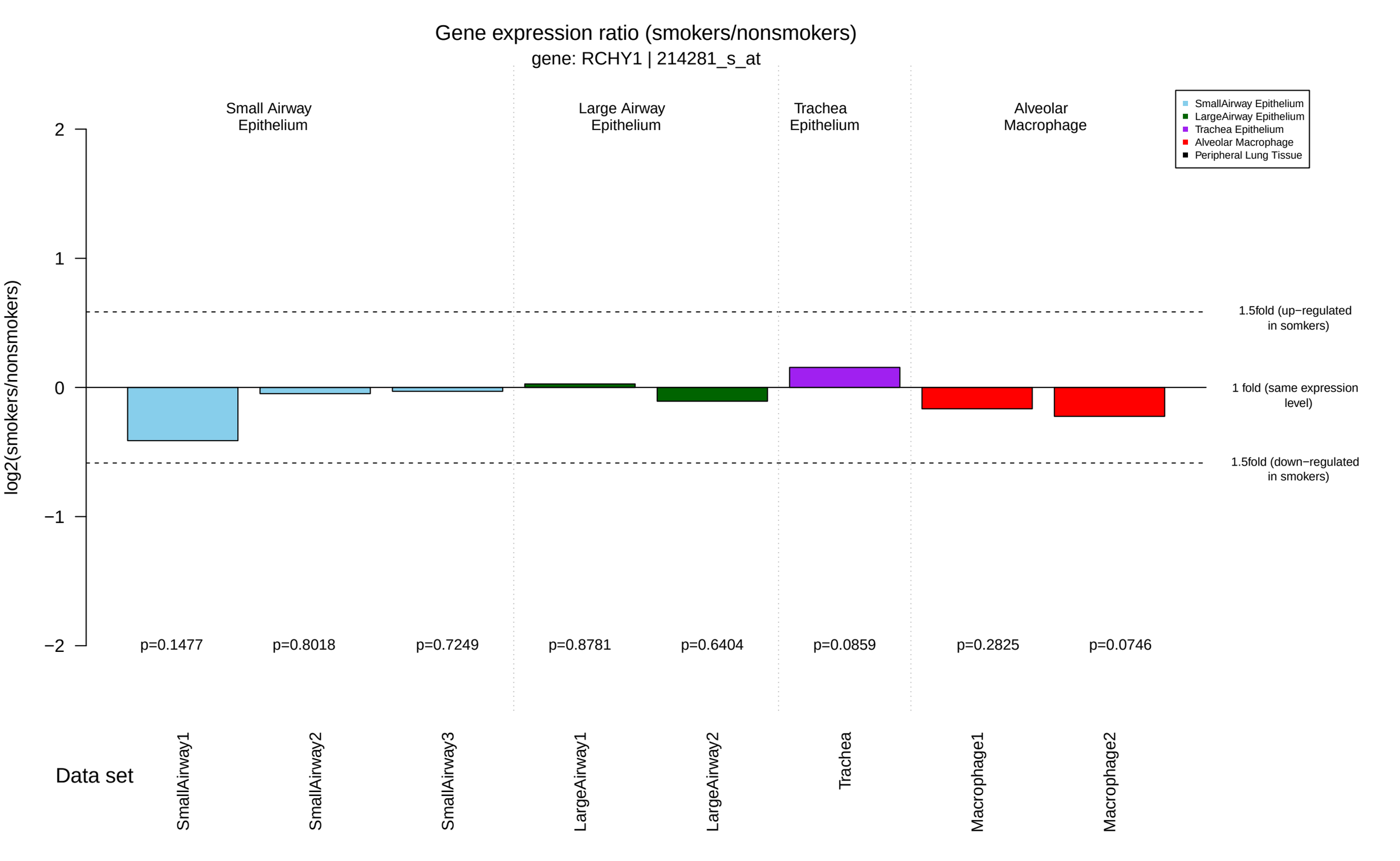
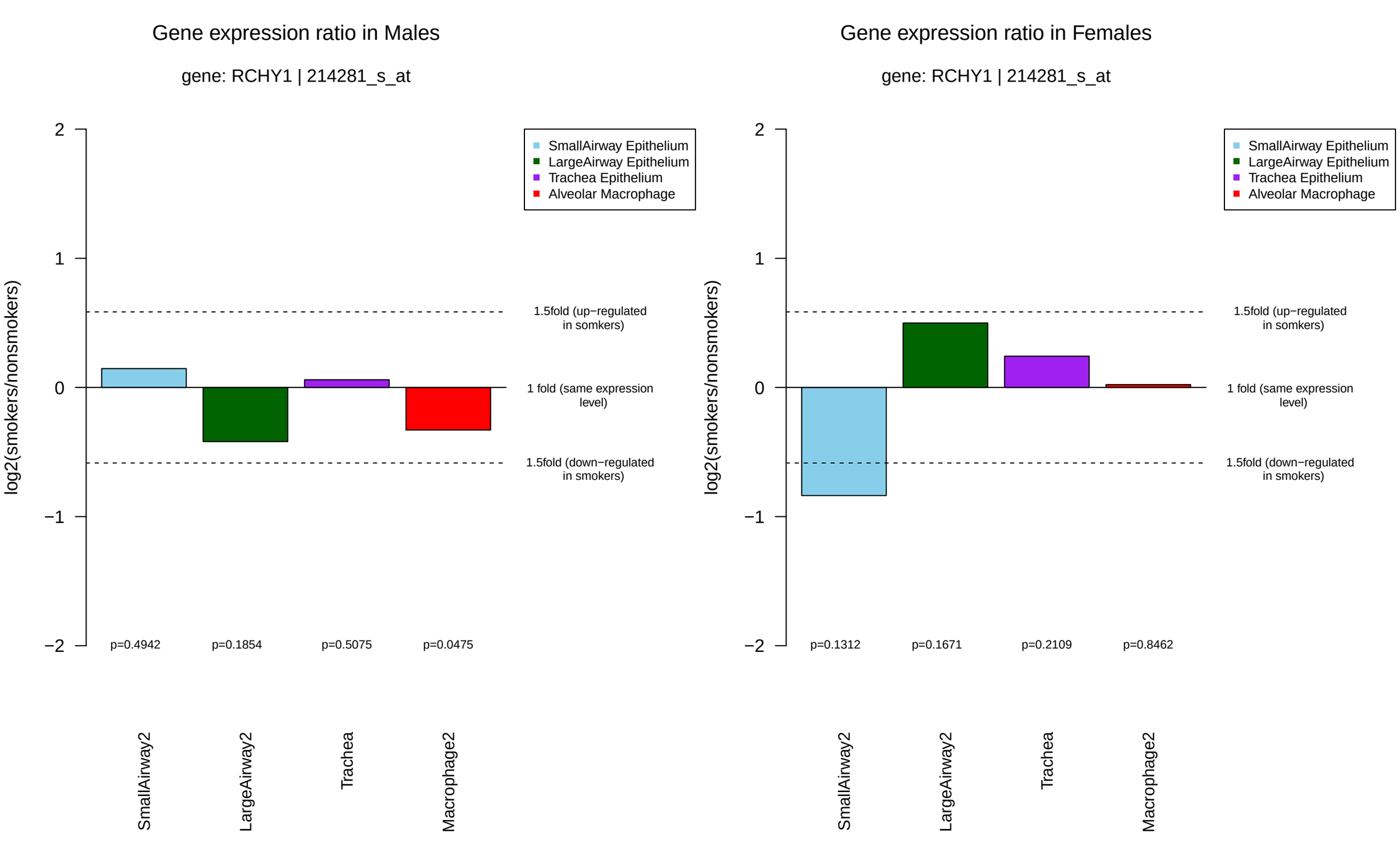
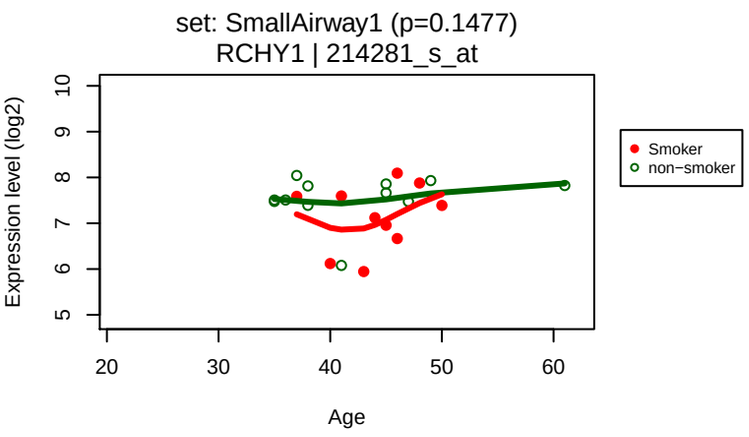
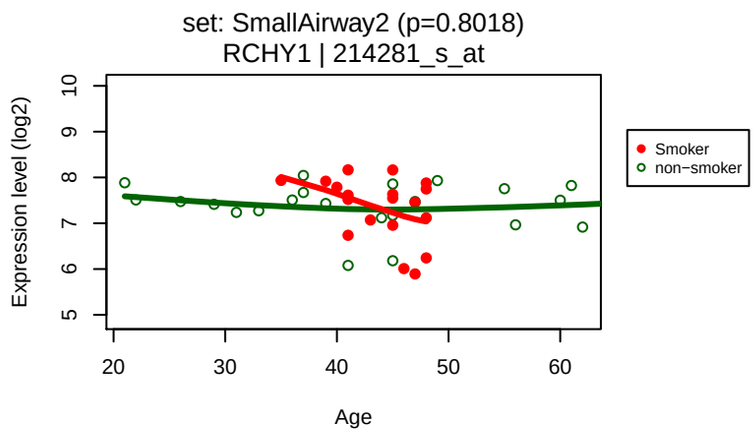
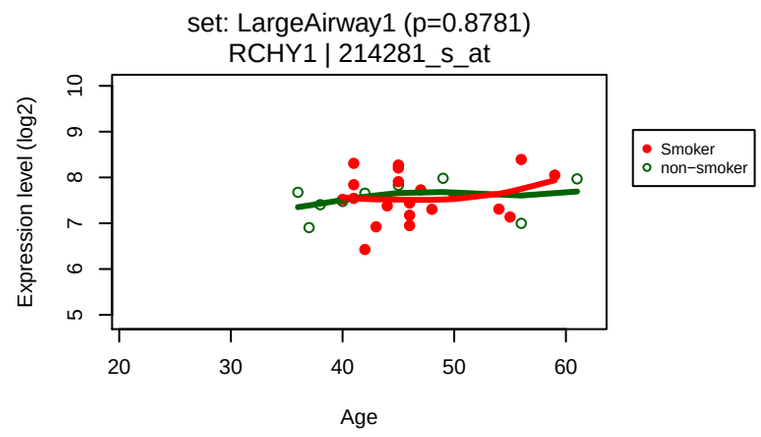

Gene expression of smokers and nonsmokers
gene: RCHY1 | 214281_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1477
p=0.8018
p=0.7249
p=0.8781
p=0.6404
p=0.0859
p=0.2825
p=0.0746
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

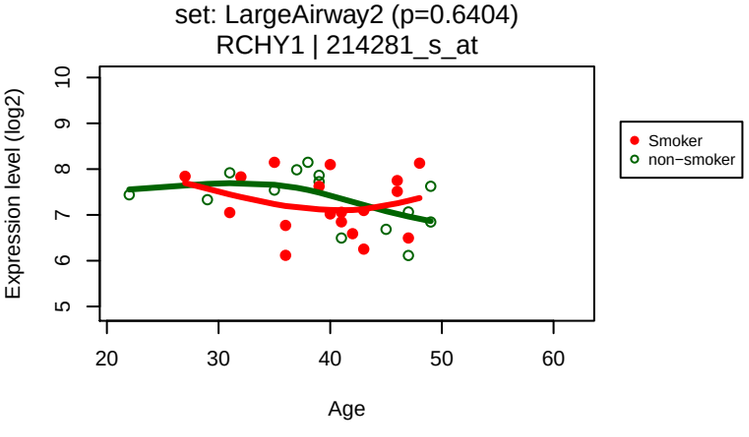

Gene expression ratio (smokers/nonsmokers)
gene: RCHY1 | 214281_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1477
p=0.8018
p=0.7249
p=0.8781
p=0.6404
p=0.0859
p=0.2825
p=0.0746
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

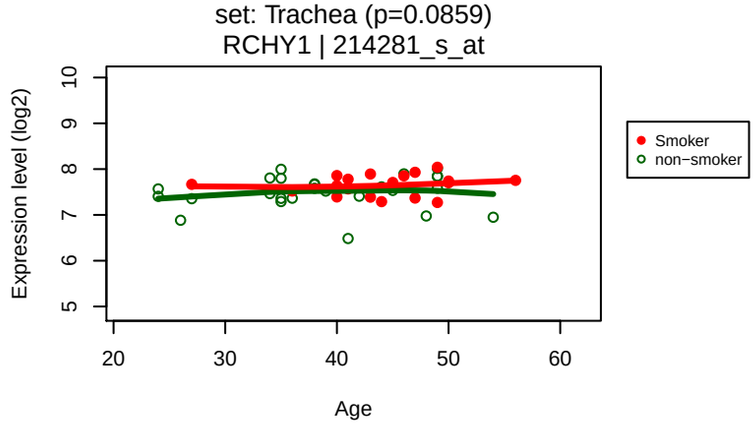

Gene expression ratio in Males
gene: RCHY1 | 214281_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4942
p=0.1854
p=0.5075
p=0.0475
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RCHY1 | 214281_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1312
p=0.1671
p=0.2109
p=0.8462
SmallAirway2
LargeAirway2
Trachea
Macrophage2

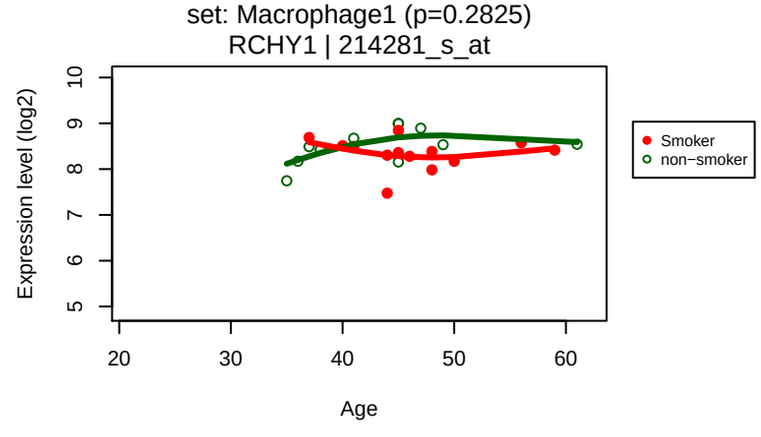

set: SmallAirway1 (p=0.1477)
RCHY1 | 214281_s_at
Expression level (log2)
Age
Smoker
non−smoker

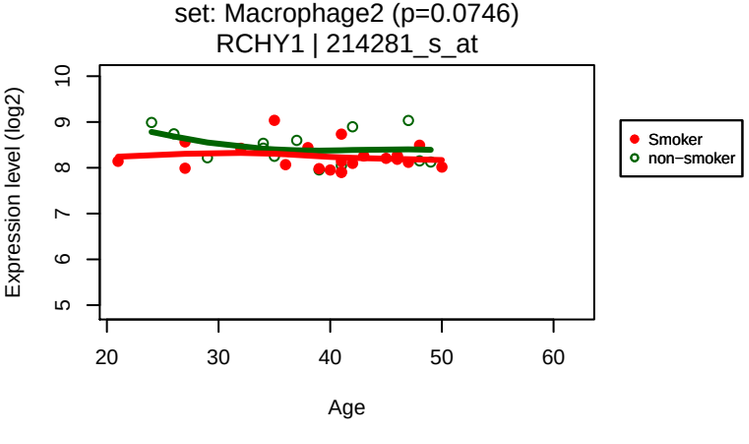

set: SmallAirway2 (p=0.8018)
RCHY1 | 214281_s_at
Expression level (log2)
Age
Smoker
non−smoker

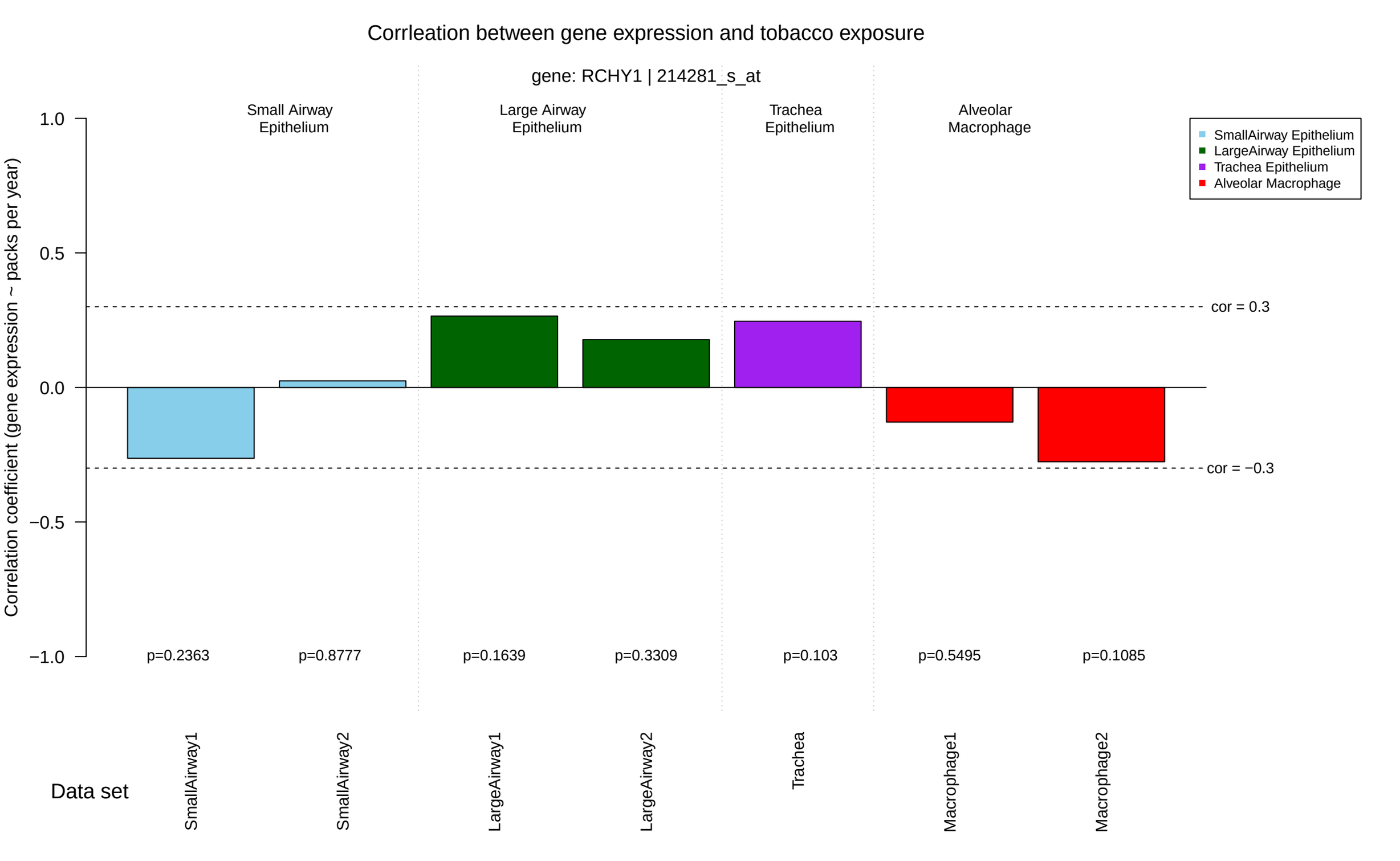

set: LargeAirway1 (p=0.8781)
RCHY1 | 214281_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6404)
RCHY1 | 214281_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0859)
RCHY1 | 214281_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2825)
RCHY1 | 214281_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0746)
RCHY1 | 214281_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RCHY1 | 214281_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2363
p=0.8777
p=0.1639
p=0.3309
p=0.103
p=0.5495
p=0.1085
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2